

Maria Ios Glarou

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🌐 meglr.github.io

Education

2018 – 2025  Integrated Master's student, School of Electrical and Computer Engineering, National Technical University of Athens (NTUA)
Major: Computer Science
Final **GPA**: 9.8/10 (10 = best, 0 = worst)
The Integrated Master's in ECE at NTUA is equivalent to the M.Sc. and M.Eng. degrees of technical universities.
Master's thesis title: *On Learning What to Learn: Adaptive Data Mixtures for Robust Multi-Target LLM Pretraining*
Jointly supervised by Prof. Martin Jaggi (EPFL) and Prof. Petros Maragos (NTUA).
Thesis conducted as part of a 7-month paid research appointment in the MLO lab at EPFL.

Research Experience

From December 2025  **EPFL** Research Assistant (Valorization of Master's Program)

- Advisor: Prof. Martin Jaggi
- Topic to be determined (direction: safety pretraining).

Spring semester of 2025  **EPFL** Research Assistant

- Advisor: Prof. Martin Jaggi
- Worked on data mixture optimization for LLM pretraining; contributed toward LLMs whose representations better support our objectives.
- Introduced GRAPE, a minimax-formulated method that can identify superior online data mixtures for multiple tasks by dynamically re-locating sampling toward lagging tasks; improved reasoning and low-resource multilingual performance over strong baselines.

Summer of 2024  **EPFL** Research Intern

- Advisor : Prof. Ola Svensson
- Explored the use of Large Language Models (LLMs) for optimizing Mixed Integer Programming (MIP) problem formulations to improve solver efficiency.
- Investigated downsizing Integer Program (IP) instances using Neural Networks by learning a smaller surrogate problem that preserves the original's objective/constraints, solving it faster, and lifting the surrogate solution back to the full instance.

Research Experience (continued)

Summer of 2023

EPFL Research Intern

- Advisor : Prof. Antoine Bosselut
- Explored methods to enhance the quality and relevance of images generated by state-of-the-art text-to-image models (Stable Diffusion). Used techniques to expand input text prompts using relevant commonsense knowledge and a lightweight transformer-based adapter to produce enhanced text embeddings and align them with the CLIP embedding used in Stable Diffusion. Conducted shortly before later releases (e.g., DALL-E 3) popularized integrated prompt expansion.

Work Experience

Summer of 2022

Hellenic Drones Machine Learning Engineer Intern

- Worked on developing a real-time system for aerial object detection and tracking by integrating YOLOv5 with DeepSORT. Achieved 87% accuracy in aerial object classification and improved tracking performance in occlusion-prone scenarios. Our model was deployed on the company's drones for aerial surveillance.

Hellenic Drones is a startup in Athens collaborating at the time with members of the Artificial Intelligence and Learning Systems (AILS) lab at NTUA.

Volunteering

Teaching Assistant

Greek National Olympiad in Informatics Workshops

- Explained algorithmic methods and example problems to high school students and assisted them during practice sessions held at NTUA.

These workshops are organized every year by Professors Dimitris Fotakis, Nikolaos Papaspyrou, and Aris Pagourtzis (NTUA).

Volunteer

Zurich AI Safety Day 2025

Research Publications

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S. Fan, M. I. Glarou, and M. Jaggi, *Grape: Optimize data mixture for group robust multi-target adaptive pretraining*, 2025. arXiv: 2505.20380 [cs.LG]. URL: <https://arxiv.org/abs/2505.20380>.

Skills

Languages



- **English:** Proficient in reading, writing, and speaking (Certificate of Competency in English, Michigan Language Institute).
- **French:** Basic understanding of spoken and written French; completed 3 years of French classes during high school.
- **Greek:** Native speaker.

Coding



Python, C++, SQL, $\text{\LaTeX}$.

Databases



Experience in MySQL, PostgreSQL, SQLite.

Web Dev



Experience in HTML, CSS, JavaScript.

Communication



Collaborated on multiple group projects at NTUA, coordinating with team members to achieve project goals on time.

Personality Traits



Curious, energetic, perspective, persistent.

Hobbies



Hiking, swimming, traveling, music, books, languages.

Misc.



Academic research.